UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2010 question paper for the guidance of teachers

5014 ENVIRONMENTAL MANAGEMENT

5014/21 Paper 2, maximum raw mark 60

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UNIVERSITY of CAMBRIDGE
International Examinations

1 (a) (i) to have control/ownership of project/eq; to learn skills/eq; could not pay for help/eq; reduce unemployment; AVP; e.g. avoid being cheated [max 2]

(ii) same number of males; and females; same area; same size/number of shelters; same food; same nest boxes; (allow same ratio/number of chickens;) [max 3]

(iii) to prevent spread of disease/eq [1]

(iv) 0.14; 0.16; allow both marks for correct working only [2]

(v) more eggs; more meat; prefer to eat it; more profit; quote relevant figures from source; faster growth rate; [max 3]

(b) (i) scale given; 6 and 80 animals indicated; two + shelters; two + water troughs; two + feeding troughs; [max 4]

(ii) supply protein/vitamins/A/D/minerals/only enough fat for health/eq; [1]

(iii) to work out expenses; to work out profit; pay wages according to hours worked; (R share profit fairly as in the stem) [2]

(iv) 150/600 × 100 = 25(%);; (allow one for correct working) [2]

(c) (i) to prevent gas leaks; to control temperature; to prevent explosions; easier to build; ref to thinner walls; less smell; ref to visual pollution; [max 1]

(ii) no need to spend time collecting wood; ref to deforestation; <u>sustainable</u> energy source; easy to light; saves money/eq; less pollution/environmentally friendly; [max 2]

(iii) charity/aid organisation (named)/government development funds/agency/eq/bank loans/ profit from sale of eggs/meat; [1]

(iv) local people produce own food; skills can be learnt/passed on to others; digestion reduces environmental damage/atmosphere qualified e.g. renewable energy source; grown on crop wastes; replace chicken stocks; [max 2]

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2 **(a)** X should be between the chicken enclosure and the village;
Valid reasons: short pipe/distance to wet rocks; short road needed; close to village for workers/eq; shorter distance to market/eq; asbestos dust less likely to harm workers/cover plants; (allow mine as source of gravel to build road/three valid reasons even if X in wrong position or not drawn) close to chickens for manure; [max 4]

(b) **(i)** Stevenson screen/wet and dry bulb/hygrometer; [1]

(ii) plants get plenty of light; for photosynthesis; high temperature; for fast growth; high humidity; limited transpiration/eq; [max 3]

(c) **(i)** 1.6 (rand) [1]

(ii) B A C;
Reasons:
shade first as seedlings would die/eq;
benches to have more pots/seedlings;
need more profit/product to justify expense for a donkey/cost of food for donkey/eq; [4]

(iii) it is a big investment/eq; extra work looking after donkey; further details e.g. vet care/ disease; cost of maintenance of cart/track; do not have/want to learn the skills needed; [max 2]

(d) **(i)** orientation; both axes labelled (days + ht in cm); plot for C; plot for D; allow one error for each plot (ignore plots for A and B) [4]

(ii) water; humidity; temp; soil type; soil pH; soil nutrients; light intensity; age of seedling; [max 2]

(iii) a fungus/damping off disease/(nematode) worms/bacteria/virus/parasite; [1]

(iv) cost; poison/contaminate themselves; other people; no protective equipment; not skilled in use; [max 2]

3 **(a)** **(i)** 800 and 275–280; [1]

(ii) 2005–2008; (allow 2 years in the range) [1]

(b) *better:* more money to pay for schooling; food; health care; household goods; seeds; livestock; tools; AVP;
R living standards unqualified
worse: men spend money before they get home(on drink); drugs; sex; may encourage crime in village; AVP;
max 2 for better or worse only [3]

(c) **(i)** 40–50 (m); plant species recover; plants per square metre recover; vegetation not likely to poison livestock; so livestock will not poison humans; [max 3]

(ii) sample beyond 50 m; more than one sample line; different times of year/eq; repeat in following years; take soil samples for cyanide; plant samples for cyanide; [max 2]

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